COOLING ENVIRONMENTS

Fluid Flush/Fill Cart for Pre-Deployment Conditioning of Liquid-Cooled IT Systems

Himanshu Modi, NVIDIA

Kevin Joseph, Meta

OCP PLATINUM Member

Uncontrolled fluid condition creates commissioning risk

CONTAMINATION
Residual fluid, particles, dissolved gas, and trapped air can restrict flow or damage components.

UNVERIFIED ACCEPTANCE
Visual checks cannot confirm cleanliness, air removal, or fluid condition.

INCONSISTENT STARTUP
PG25 concentration and handling vary by supplier and site, increasing rework and time to acceptance.

NO AUDIT TRAIL
Without a common telemetry schema and cycle record, results cannot be compared or reproduced.

Operational consequence: longer per-rack commissioning, repeat flush/fill cycles, and avoidable component risk.
Target before service: clean, air-free, in-spec PG25 verified by measured acceptance criteria.

Standardize the interface, not the cart design

The Base Specification defines operating envelopes and interfaces. Pump topology, controls, and packaging remain open to suppliers.

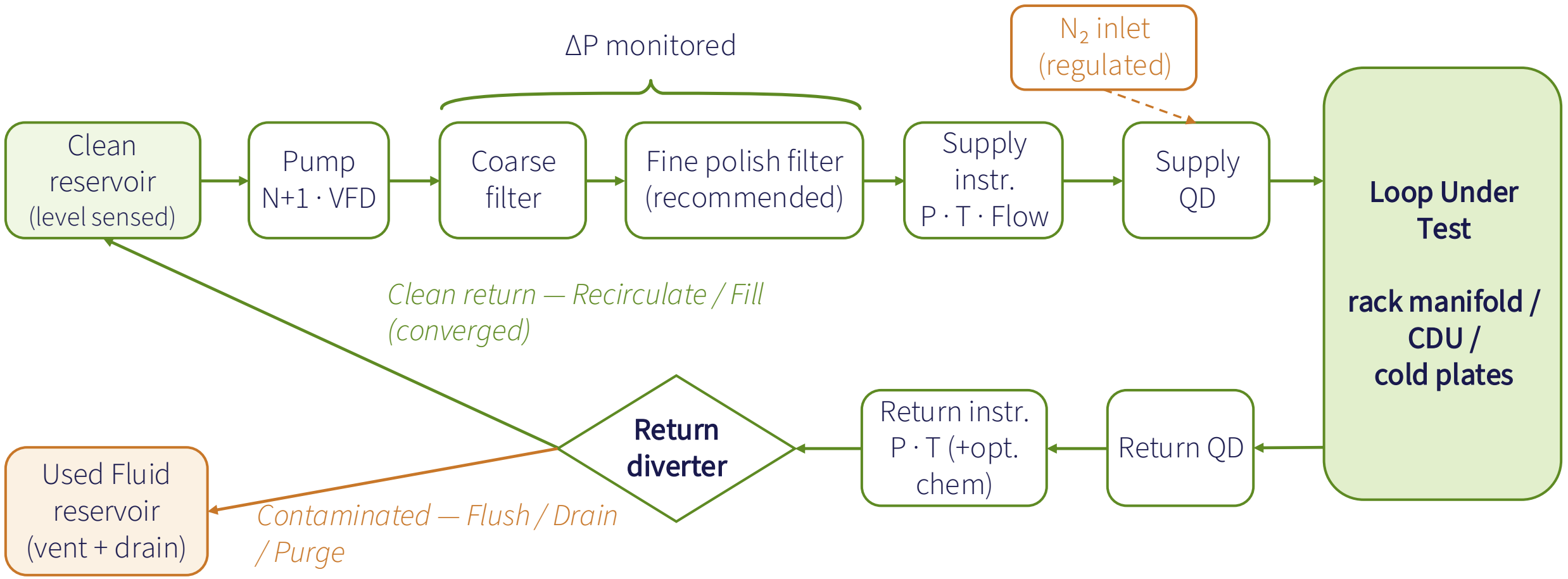
IN

- Mobile, single-operator cart
- PG25 on CDU-to-rack TCS loops
- Supply + return dry-break interfaces
- Regulated nitrogen inlet
- Flush, fill, condition, drain, sample, and dry

OUT

- Facility-water handling or treatment
- Two-phase or dielectric working fluids
- Permanently installed CDUs
- Product-level pump, control, or packaging design

Clean and recovered fluid never mix



CONDITION • ROUTE • RECORD

Six modes cover the loop lifecycle

Flush	**One-way fill**	**Closed-loop condition**	**Drain**	**Sample**	**N_2 Purge/Dry**

Required Process **Recommended Process**

Six modes cover the loop lifecycle

Flush	One-way fill	Closed-loop condition	Drain	Sample	N_2 Purge/Dry
Remove residue Route return to recovery	Displace air without mixing returns	Recirculate through filters to acceptance	Recover bulk fluid from the loop	Capture a controlled quality sample	Remove residual fluid and moisture

Required Process **Recommended Process**

From improvised rigs to interchangeable tools

Dimension	Typical today	With a common base spec
Fluid handling	Single reservoir; cross-contamination risk	Separate clean and recovered paths
Acceptance	Visual inspection	Measured acceptance criteria
Traceability	Limited or no cycle history	Published telemetry and auditable cycle logs
Protection	Site-specific safeguards	Defined fault responses, security controls, and containment

A common base specification also standardizes six operating modes and filter differential-pressure monitoring.

Three deployment interfaces to get right

MOBILITY

Single-operator movement

Site access without disassembly

Stable, maneuverable, and serviceable

POWER

Regional power-input compatibility

Electrical leakage detection recommended

UPS recommended for mobile operation

COUPLINGS

Readily available sanitary flanges and QDs

Tested and validated for PG25 service

Sized to loop demand

Clearly labeled to prevent misconnection

Instrumented. Interlocked. Fluid-compatible.

INSTRUMENTED

Published data interface

Time-synchronized process and equipment-state data

Exportable cycle records for acceptance and traceability

INTERLOCKED

Leak, overpressure, O_2, and phase faults

Defined inhibit, isolation, and mode-abort actions

Critical alarms cannot be suppressed

FLUID-COMPATIBLE

Passivated 304L/316L and PTFE/PEEK-class materials

Peroxide-cured EPDM or FKM seals

ISO 4406 18/16/13 cleanliness

Call to Action

- Next Steps:
  - Collect feedback from the broader community.
  - Cooling Environments committee submission and approval
  - Provide future amendments and feedback on the published specifications

- Join the OCP Cooling Environments workstream

OCP Cooling Environments

Thank You!

